Problem 1. Consider the function $d\colon \mathbb{N} \to \mathbb{N}$ defined $d(n) =$ the number of divisors of n. For example, $d(12) = 6$ (because the divisors of 12 are $1, 2, 3, 4, 6, 12$).

1. Let A be the set of all natural numbers n such that $d(n) = 2$. (I.e., A is the *preimage* of 2.) Determine what A is.

   Solution. Note that that, for every positive integer a, we have $a \in A$ if and only if a has exactly two divisors. We claim that A is the set of prime numbers. We prove both inclusions.

   By the definition of prime numbers (from the first day of class!) a number a is prime if it cannot be written as a product of two smaller natural numbers. This means that a prime number a has only two divisors, 1 and a (any other divisor would lead to a factorization of a into two smaller numbers). So certainly the set of prime numbers is a subset of A.

   Conversely, if a number a has precisely two divisors, then we know $a = 1 \cdot a$ so 1 and a are the only divisors of a. This means that we cannot write a as the product of two smaller numbers, so a is prime.

   Therefore, A is exactly the set of prime numbers. □

---

2. Prove or disprove the following statement.

   $$\text{Let } m, n \in \mathbb{N}. \text{ If } m \neq n, \text{ then } d(m) \neq d(n).$$

   Solution. This is false. Notice that $d(2) = d(3) = 2$ by part (1), but $2 \neq 3$. □

---

3. Let $n \in \mathbb{N}$. Prove that if a natural number a divides 2^{n+1}, then either $a = 2^{n+1}$ or a divides 2^n.

   Proof. We will prove this by induction on n.

   Base Case. Let $n = 1$. Then $2^{n+1} = 2^2$. If a divides 4 with $a \neq 4$, then $a = 2$ or $a = 1$, so a divides $2^n = 2^1$. Thus the statement holds for $n = 1$.

   Inductive Step. Let $k \in \mathbb{N}$ and assume that the statement holds for k (i.e., if a divides 2^{k+1}, then either $a = 2^{k+1}$ or a divides 2^k.) We will show that it holds for $k + 1$.

   Assume that a divides 2^{k+2}, and assume that $a \neq 2^{k+2}$. If we can show that a divides 2^{k+1} then we will be done. Since a divides 2^{k+2}, we know that

   $$2^{k+2} = ab$$

   for some $b \in \mathbb{Z}_{\geqslant 0}$. Since $a \neq 2^{k+2}$, thus $b > 1$. Since ab is even, we know by a theorem from class that either a is even or b is even.

   Case I. Assume that a is even. Thus $a = 2a'$ for some $a' \in \mathbb{Z}$. Therefore

   $$2^{k+2} = ab = 2a'b,$$

   so by cancellation, $2^{k+1} = a'b$. By our induction hypothesis, either $a' = 2^{k+1}$ or a' divides 2^k. If $a' = 2^{k+1}$, then this contradicts our assumption (why?), so a' divides 2^k. Thus $a = 2a'$ divides 2^{k+1}. Therefore this case is complete.

   Case II. Assume that b is even. Thus $b = 2b'$ for some $b' \in \mathbb{Z}$. Therefore

   $$2^{k+2} = ab = 2ab',$$

   so by cancellation, $2^{k+1} = ab'$. Thus a divides 2^{k+1}.

   This completes our proof by induction. Thus the lemma is true. □

---

4. Find a formula for $d(2^n)$ for all $n \in \mathbb{N}$ and prove that your formula is correct.

   Solution. We claim that for $n \in \mathbb{N}$, we have $d(2^n) = n + 1$.

   Now we prove our claim by induction on n.

   Base Case. When $n = 1$, then, using that 2 is prime, from part (1) we have $d(2) = 2$ so $d(2^1) = 1 + 1$.

   Inductive Step. Suppose that for some $k \in \mathbb{N}$, we have $d(2^k) = k + 1$. We want to know the number of divisors of 2^{k+1}.

   But by the lemma, if a is a divisor of 2^{k+1}, then $a = 2^{k+1}$ or a is a divisor of 2^k. Hence, the possible divisors of 2^{k+1} are 2^{k+1} itself, or any of the $k + 1$ divisors of 2^k.

   Since $2^{k+1} > 2^k$, it is clear that 2^{k+1} is not one of the $k + 1$ divisors of 2^k. Hence, there are exactly $k + 2$ divisors of 2^{k+1}, in other words $d(2^{k+1}) = k + 2$.

   By induction, we have proved the claim. □

5. Is d a surjective function? (Prove your answer.)

   Solution. We show that d is surjective. Let $m \in \mathbb{N}$. Then $2^{m-1} \in \mathbb{N}$ is such that $d(2^{m-1}) = m$. □

Problem 2. Let $M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \;\middle|\; a, b, c, d \in \mathbb{R} \right\}$, i.e. M_2 is the set of 2×2 matrices with entries in $\mathbb{R}$. We define a function $T\colon M_2 \to M_2$ by

$$T\left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right\} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

We call T the **transpose** function. Answer the following, and justify your answers completely.

1. Is T injective?

   Solution. The function T is injective, and here's a proof.

   Let $A, B \in M_2$ such that $T(A) = T(B)$. By definition of M_2, there exist real numbers a, b, c, d, e, f, g, h such that $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$. then $T(A) = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ and $T(B) = \begin{bmatrix} e & g \\ f & h \end{bmatrix}$. Since $T(A) = T(B)$, thus

   $$\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} e & g \\ f & h \end{bmatrix}.$$

   Therefore $a = e$, $b = f$, $c = g$, and $d = h$. In other words, $A = B$. Thus T is injective. □

---

2. Is T surjective? **Solution.** The function T is surjective and here's a proof.

   Let $A \in M_2$. By definition of M_2, this means there exist $a, b, c, d \in \mathbb{R}$ such that $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

   $$T\left(\begin{bmatrix} a & c \\ b & d \end{bmatrix} \right) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = A,$$

   so T is surjective. □

---

3. Is T bijective?

   Solution. A function is bijective if and only if it is both injective and surjective. So by Parts (1) and (2), we see that T is bijective. □

---

4. Let $S \subseteq M_2$ be the set such that $T(X) = X$ for all $X \in S$. Describe S as explicitly as possible.

   Solution. I claim that

$$S = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\}. \tag{1}$$

   This set is called the set of *symmetric* matrices. I will show that Equation (1) holds by showing that each set is a subset of the other.

   Let $X \in \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\}$. Thus $X = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ for some $a, b, c \in \mathbb{R}$, so $T(X) = \begin{bmatrix} a & b \\ b & c \end{bmatrix} = X$. Therefore $X \in S$.

   Conversely, assume that $X \in S$. Thus $T(X) = X$. So if $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, this means that

$$\begin{bmatrix} x_1 & x_3 \\ x_2 & x_4 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}.$$

   Thus $x_2 = x_3$, which means that X is a symmetric matrix. This completes our proof. □

---

5. Let $U = \left\{ \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} \;\middle|\; a, b, d \in \mathbb{R} \right\}$. Describe $T^{-1}(U)$ as explicitly as possible.

   Solution. I claim that $T^{-1}(U) = L$ where

$$L = \left\{ \begin{bmatrix} a & 0 \\ b & d \end{bmatrix} \right\}.$$

   By the way, U stands for "upper triangular matrices" and L stands for "lower triangular matrices."

   The above equality is more or less immediate: If $X \in L$, then $T(X) \in U$, so $X \in T^{-1}(U)$. If instead $X \in T^{-1}(U)$, then this shows that $X \in L$. This completes the proof. □

Problem 3. Assume A and B are nonempty sets and let $f\colon A \to B$. Prove or disprove the following.

1. For all $U \subseteq A$ and $V \subseteq A$, $f(U \cup V) = f(U) \cup f(V)$.

   Solution. This statement appears to be true, so let's prove it. Let A and B be sets, $U, V \subseteq A$, and $f : A \to B$ be a function. We will show this equality by showing the two sets are subsets of each other.

   Recall that if $W \subset A$, then

$$f(W) = \{b \in B \mid b = f(a) \text{ for some } a \in W\}.$$

   Let $x \in f(U \cup V)$. Thus $x = f(a)$ for some $a \in U \cup V$. If $a \in U$, then $x = f(a) \in f(U)$. If $a \in V$, then $x = f(a) \in f(V)$. In either case, $x \in f(U) \cup f(V)$. Therefore $f(U \cup V) \subseteq f(U) \cup f(V)$.

   Instead let $x \in f(U) \cup f(V)$. Without loss of generality, assume that $x \in f(U)$ (the other case will proceed precisely the same, replacing U with V). Thus $x = f(a)$ for some $a \in U$. Thus $a \in U \cup V$, so $x = f(a) \in f(U \cup V)$. Therefore $f(U) \cup V \subseteq f(U \cup V)$. This completes the proof. □

---

2. For all $X \subseteq B$ and $Y \subseteq B$, $f^{-1}(X \cap Y) = f^{-1}(X) \cap f^{-1}(Y)$.

   Solution. This statement is also true.

   Let A and B be sets, $X, Y \subseteq B$, and $f : A \to B$ be a function. We'll again show that the two sets in question are subsets of each other.

   Recall that if $W \subseteq B$, then
   $$f^{-1}(W) = \{a \in A \mid f(a) \in W\}.$$

   Let $a \in f^{-1}(X \cap Y)$. Thus $f(a) \in X \cap Y$. By definition of intersection, $f(a) \in X$ and $f(a) \in Y$. Thus $a \in f^{-1}(X)$ and $a \in f^{-1}(Y)$. Therefore $a \in f^{-1}(X) \cap f^{-1}(Y)$. Therefore $f^{-1}(X \cap Y) \subseteq f^{-1}(X) \cap f^{-1}(Y)$.

   Instead assume that $a \in f^{-1}(X) \cap f^{-1}(Y)$. Thus $a \in f^{-1}(X)$ and $a \in f^{-1}(Y)$. This means that $f(a) \in X$ and $f(a) \in Y$. Therefore $f(a) \in X \cap Y$, so $a \in f^{-1}(X \cap Y)$. Thus $f^{-1}(X) \cap f^{-1}(Y) \subseteq f^{-1}(X \cap Y)$. Therefore the two sets are equal, so our proof is complete. □

Problem 4. Let A, B, and C be nonempty sets and $f\colon A \to B$.

1. Assume that f is surjective. Show that if $g\colon B \to C$ and $h\colon B \to C$ are functions such that $g \circ f = h \circ f$, then $g = h$. **Proof.** Let A, B, and C be nonempty sets, $f\colon A \to B$, and $g, h\colon B \to C$. Assume that f is surjective and that $g \circ f = h \circ f$. We will show that $g = h$.

   Recall that two functions g and h are equal if and only if they have the same domain and codomain and $g(x) = h(x)$ for all x in their domain. Since g and h already have the same domain and codomain, all we need to show is that $g(x) = h(x)$ for all $x \in B$.

   We will prove this by contradiction. Assume there exists an $x \in B$ such that $g(x) \neq h(x)$. Since f is surjective, there exists an $a \in A$ such that $f(a) = x$. Thus $g(x) = g(f(a)) = (g \circ f)(a)$ and $h(x) = h(f(a)) = (h \circ f)(a)$. Thus $(g \circ f)(a) \neq (h \circ f)(a)$, but this contradicts the assumption that $g \circ f = h \circ f$. Therefore no such x can exist, and thus $g = h$. □

---

2. Show that Part 1 is not necessarily true if f is not surjective.

   Solution. Consider $A = \{1\}$, $B = \{2, 3\}$ and $C = \{4, 5\}$. Define $f\colon A \to B$ as $f(1) = 2$. Define $g : B \to C$ as $g(2) = 4$, $g(3) = 5$, and define $h : B \to C$ as $h(2) = 4$, $h(3) = 4$. Then $g \circ f = h \circ f$ but $g \neq h$. □

Problem 5. Let $f\colon A \to B$ and $g\colon B \to C$. Prove that if $g \circ f$ is a surjection, then g is a surjection. Give a counterexample to show that the converse does not hold.

Solution. Let $f\colon A \to B$ and $g\colon B \to C$ and that $g \circ f\colon A \to C$ is a surjection. We wish to show that g is a surjection.

To do this, let $c \in C$. We must find some $b \in B$ such that $g(b) = c$. Since $g \circ f$ is surjective, we know there exists $a \in A$ such that $(g \circ f)(a) = c$. Let $b = f(a) \in B$.

Then $g(b) = g(f(a)) = (g \circ f)(a) = c$. Hence, we have shown that g is surjective.

To show that the converse does not hold, let $A = \{a, b\}$, $B = \{p, q\}$ and $C = \{x, y\}$. Define $g\colon B \to C$ by $g(p) = x$ and $g(q) = y$, which is clearly a surjection. Define $f\colon A \to B$ by $f(a) = f(b) = p$. Then $g \circ f\colon A \to C$ is not a surjection since $(g \circ f)(a) = x$ and $(g \circ f)(b) = x$, so y is not in the image of $g \circ f$. □

Problem 6. Let $f\colon A \to B$ and $g\colon B \to C$. Prove that if f and g are both injective, then $g \circ f$ is injective. Give a counterexample to show that the converse does not hold.

Solution. Suppose that $f\colon A \to B$ and $g\colon B \to C$ are both injective. We wish to show that $g \circ f\colon A \to C$ is injective. So suppose that there exist $a, b \in A$ such that $(g \circ f)(a) = (g \circ f)(b)$. We must show that $a = b$. By the definition of composition, we have $g(f(a)) = g(f(b))$. Since g is injective, this means that $f(a) = f(b)$. Now since f is injective, we conclude $a = b$. Altogether, we have shown that $g \circ f$ is injective.

To show that the converse does not hold, let $A = \{a\}$, $B = \{p, q\}$ and $C = \{x\}$. Define $f(a) = p$, $g(p) = g(q) = x$. Then the composition $g \circ f$ is injective, but g is not injective, because $p \neq q$ but $g(p) = g(q)$. □

Problem 7. Do not turn these in, but complete the progress checks in the sections of textbook.